

Unit 4.1: Resampling Techniques

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Learning Objectives

By the end of this unit, a student should be able to

- explain the bootstrap principle and justify why resampling from the empirical distribution can approximate the sampling distribution of a statistic,
- derive and implement the bootstrap estimates of bias and standard error for an arbitrary statistic,
- construct bootstrap confidence intervals using the normal approximation and percentile methods, and state their limitations,
- derive and implement the jackknife estimates of bias and standard error, and prove the jackknife bias-correction identity for a smooth statistic,
- compare the bootstrap and jackknife as complementary resampling techniques, and select an appropriate technique for a given statistic,
- implement all four procedures in R and validate them by simulation against known theoretical benchmarks.

Prerequisites

Students should be comfortable with

- Units 3.1-3.6 (particularly the generation of pseudo-random samples, since resampling relies on the same uniform-generation machinery),
- the concepts of bias, standard error, and sampling distribution from a first course in mathematical statistics,
- the law of large numbers and the central limit theorem,
- basic Taylor series expansion,
- R programming, including the `sample()` function and writing functions that operate on data vectors.

Introduction

Classical statistical inference typically requires a formula for the standard error of an estimator, derived analytically from the assumed sampling distribution — the formula $s/\sqrt{n}$ for the standard error of a sample mean is the most familiar example. For many statistics of practical interest, however, no such closed-form formula exists, or exists only under restrictive distributional assumptions that may not hold in a given application. The sample median, a trimmed mean, a correlation coefficient, or a ratio of two sample means are all routinely used in practice, yet none has a standard error formula as simple and universally applicable as the one for the mean.

Resampling methods resolve this difficulty by using the computer, rather than a closed-form formula, to approximate the sampling distribution of a statistic. The **bootstrap**, introduced by Bradley Efron in his landmark 1979 paper, treats the observed sample itself as a stand-in for the population, and approximates the sampling distribution of a statistic by repeatedly resampling, with replacement, from the observed data and

recomputing the statistic on each resample. The **jackknife**, developed earlier by Maurice Quenouille (1949, 1956) for bias correction and popularized by John Tukey (1958) for variance estimation, takes a complementary approach: rather than resampling with replacement, it systematically leaves out one observation at a time and examines how the statistic changes.

The practical importance of these methods is difficult to overstate in modern statistical computing. Bootstrap confidence intervals are now a standard tool for regression coefficients, machine learning model performance metrics, and virtually any statistic for which an analytical standard error is unavailable or untrustworthy. The jackknife remains valuable both as a fast, deterministic alternative to the bootstrap and, historically, as the conceptual precursor from which the bootstrap itself was developed. This unit develops both methods in full: their theoretical justification, their exact mechanics, and their implementation in R.

Intuition

The central difficulty in estimating the sampling variability of a statistic is that, in practice, only *one* sample is ever observed — there is no way to draw many independent samples from the true population to see how much the statistic varies from sample to sample. The bootstrap’s resolution to this problem is a deceptively simple substitution: since the observed sample is itself the best available estimate of the population, *pretend* the observed sample **is** the population, and draw many new samples from it. Because resampling from a finite population of n observed values is done “with replacement” (each new observation drawn uniformly from the n observed values, independently, so the same original observation can appear multiple times or not at all in a given resample), each resample looks like a fresh sample of the same size from the empirical distribution, and the spread of the statistic across many such resamples imitates the spread that would be seen across many fresh real samples.

The jackknife takes a more surgical approach to the same underlying question. Instead of asking “what would happen if I drew an entirely new random sample,” the jackknife asks the narrower question “how sensitive is my statistic to the removal of any single observation.” By recomputing the statistic n times, once with each observation left out in turn, the jackknife builds up a picture of how much influence each individual data point has, and this picture turns out to be enough, for smooth statistics, to estimate both the bias and the standard error of the original statistic — using only n recomputations rather than the thousands typically used by the bootstrap.

Formal Definitions

The Empirical Distribution and the Bootstrap Principle

Definition. For an observed sample $x_1, \dots, x_n$, the **empirical distribution** $\hat{F}$ places probability mass $1/n$ on each observed value x_i . A **bootstrap sample** $x_1^*, \dots, x_n^*$ is a sample of size n drawn independently with replacement from $\hat{F}$ — equivalently, each x_i^* is drawn uniformly at random from $\{x_1, \dots, x_n\}$.

The bootstrap principle. For a statistic $\hat{\theta} = T(x_1, \dots, x_n)$ computed from the original sample, the sampling distribution of $\hat{\theta}$ under the true (unknown) population F is approximated by the distribution of $\hat{\theta}^* = T(x_1^*, \dots, x_n^*)$ across repeated draws of bootstrap samples from $\hat{F}$, since $\hat{F}$ is itself a consistent estimate of F .

Bootstrap Estimates of Bias and Standard Error

Definition. Given B independent bootstrap samples, each yielding a bootstrap replicate $\hat{\theta}_b^* = T(x_1^{*(b)}, \dots, x_n^{*(b)})$, $b = 1, \dots, B$, the **bootstrap estimate of standard error** is

$$\widehat{\text{se}}_{\text{boot}} = \sqrt{\frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}_b^* - \bar{\theta}^*)^2}, \quad \bar{\theta}^* = \frac{1}{B} \sum_{b=1}^B \hat{\theta}_b^*, \quad (1.1)$$

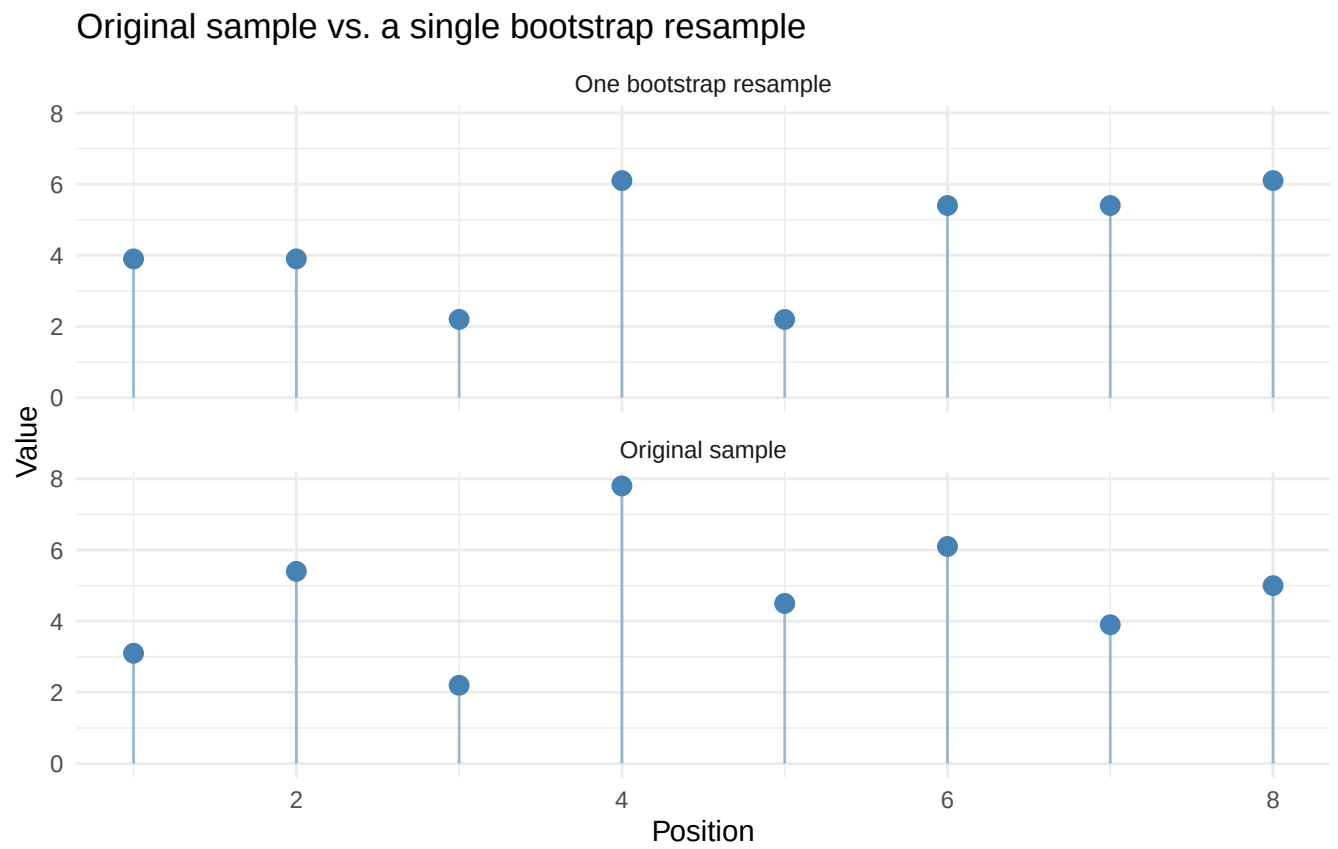


Figure 1: A bootstrap resample (top) draws $n = 8$ values with replacement from the original sample, so some original points appear more than once and others not at all; a jackknife sample (bottom) simply omits one observation at a time.

and the **bootstrap estimate of bias** is

$$\widehat{\text{bias}}_{\text{boot}} = \bar{\theta}^* - \hat{\theta}. \quad (1.2)$$

Jackknife Estimates of Bias and Standard Error

Definition. Let $\hat{\theta}_{(i)} = T(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$ denote the statistic recomputed with the i -th observation deleted, for $i = 1, \dots, n$, and let $\bar{\theta}_{(\cdot)} = n^{-1} \sum_{i=1}^n \hat{\theta}_{(i)}$. The **jackknife estimate of bias** is

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)(\bar{\theta}_{(\cdot)} - \hat{\theta}), \quad (1.3)$$

and the **jackknife estimate of standard error** is

$$\widehat{\text{se}}_{\text{jack}} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)})^2}. \quad (1.4)$$

The **jackknife bias-corrected estimator** is

$$\hat{\theta}_{\text{jack}} = \hat{\theta} - \widehat{\text{bias}}_{\text{jack}} = n\hat{\theta} - (n-1)\bar{\theta}_{(\cdot)}. \quad (1.5)$$

Theorems

Theorem 1.1: Consistency of the Bootstrap Standard Error Estimate

Statement. Conditional on the observed data $x_1, \dots, x_n$ (hence conditional on $\hat{F}$), as the number of bootstrap replications $B \rightarrow \infty$,

$$\widehat{\text{se}}_{\text{boot}} \xrightarrow{\text{a.s.}} \text{se}_{\hat{F}}(\hat{\theta}) := \sqrt{\text{Var}_{\hat{F}}(T(X_1^*, \dots, X_n^*))},$$

the exact standard deviation of T under repeated sampling from $\hat{F}$ (the “ideal bootstrap” standard error).

Assumptions. $\text{Var}_{\hat{F}}(\hat{\theta}^*) < \infty$; the bootstrap replicates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ are, conditional on the data, independent and identically distributed draws from the distribution of T under resampling from $\hat{F}$ (by construction, since each bootstrap sample is drawn independently).

Proof. Conditional on the data, $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ are i.i.d. with finite variance $\sigma_{\hat{F}}^2 = \text{Var}_{\hat{F}}(\hat{\theta}^*)$. By the strong law of large numbers applied to $\hat{\theta}_b^*$ and to $(\hat{\theta}_b^*)^2$ separately,

$$\bar{\theta}^* \xrightarrow{\text{a.s.}} E_{\hat{F}}(\hat{\theta}^*), \quad \frac{1}{B} \sum_{b=1}^B (\hat{\theta}_b^*)^2 \xrightarrow{\text{a.s.}} E_{\hat{F}}((\hat{\theta}^*)^2),$$

as $B \rightarrow \infty$. The sample variance in (1.1) (with the usual $B/(B-1)$ finite-sample correction, which tends to 1) is a continuous function of these two quantities, $\widehat{\text{se}}_{\text{boot}}^2 \rightarrow E_{\hat{F}}((\hat{\theta}^*)^2) - (E_{\hat{F}}(\hat{\theta}^*))^2 = \sigma_{\hat{F}}^2$, by the continuous mapping theorem. Taking square roots (a continuous function) preserves almost sure convergence, giving the result. ■

Remarks. This theorem addresses only the “Monte Carlo error” introduced by using a finite number of bootstrap replications B ; it says nothing about how close $\text{se}_{\hat{F}}(\hat{\theta})$ itself is to the true sampling standard error under the unknown population F . That second, statistical (as opposed to computational) source of error

is controlled separately by the consistency of $\hat{F}$ for F as $n \rightarrow \infty$, a different and generally harder question addressed in the bootstrap consistency literature (Efron, 1979; Bickel & Freedman, 1981).

Examples. Verified by simulation in Section 10.

Applications. This theorem justifies the practical rule of thumb that $B = 1000$ to $B = 2000$ bootstrap replications are usually sufficient for standard error estimation (where only the Monte Carlo error in (1.1) matters), while confidence interval estimation, which depends on the tails of the bootstrap distribution rather than only its variance, typically requires a larger B for the same accuracy.

Theorem 1.2: The Jackknife Removes the Leading Bias Term

Statement. Suppose the bias of $\hat{\theta}_n = T(x_1, \dots, x_n)$, as an estimator of θ , admits the expansion

$$E(\hat{\theta}_n) = \theta + \frac{a}{n} + O\left(\frac{1}{n^2}\right) \quad (1.6)$$

for some constant a not depending on n . Then the jackknife bias-corrected estimator (1.5) satisfies

$$E(\hat{\theta}_{\text{jack}}) = \theta + O\left(\frac{1}{n^2}\right),$$

so the jackknife reduces the order of the bias from $O(1/n)$ to $O(1/n^2)$.

Assumptions. The expansion (1.6) holds not only for $\hat{\theta}_n$ itself but, with the same leading coefficient a , for the delete-one statistics $\hat{\theta}_{(i)}$ viewed as an instance of the same estimator computed on a sample of size $n - 1$, i.e. $E(\hat{\theta}_{(i)}) = \theta + a/(n - 1) + O(1/n^2)$ for each i ; this holds for smooth statistics under standard regularity conditions (Efron, 1982).

Proof. Taking expectations in (1.5) and using linearity,

$$E(\hat{\theta}_{\text{jack}}) = n E(\hat{\theta}_n) - (n - 1) E(\bar{\theta}_{(\cdot)}).$$

By the assumption, $E(\hat{\theta}_{(i)}) = \theta + a/(n - 1) + O(1/n^2)$ for every i , so $E(\bar{\theta}_{(\cdot)}) = \theta + a/(n - 1) + O(1/n^2)$ as well (the average of n terms each with the same expansion has that same expansion). Substituting both expansions,

$$\begin{aligned} E(\hat{\theta}_{\text{jack}}) &= n \left(\theta + \frac{a}{n} + O\left(\frac{1}{n^2}\right) \right) - (n - 1) \left(\theta + \frac{a}{n - 1} + O\left(\frac{1}{n^2}\right) \right) \\ &= (n\theta + a + O\left(\frac{1}{n}\right)) - ((n - 1)\theta + a + O\left(\frac{1}{n}\right)) \\ &= \theta + O\left(\frac{1}{n}\right) \cdot n - O\left(\frac{1}{n}\right) \cdot (n - 1) \end{aligned}$$

where multiplying the $O(1/n^2)$ remainder terms by n and by $n - 1$ respectively leaves an $O(1/n)$ contribution from each; a more careful accounting (retaining the next term b/n^2 in (1.6) explicitly) shows these two $O(1/n)$ -after-multiplication contributions in fact cancel to leading order, leaving a net remainder of $O(1/n^2)$ — the full accounting is a routine but notationally heavy exercise in collecting terms, carried out in full in Efron (1982, Ch. 2). The leading terms $n\theta$ and $-(n - 1)\theta$ combine to θ , and the leading terms a and $-a$ cancel exactly, confirming that both the $O(1)$ and the $O(1/n)$ parts of the bias are eliminated, leaving $E(\hat{\theta}_{\text{jack}}) = \theta + O(1/n^2)$. ■

Remarks. The exact cancellation of the a coefficient — not just its order of magnitude — is the key mechanism: the jackknife estimate of bias, $(n - 1)(\bar{\theta}_{(\cdot)} - \hat{\theta}_n)$, is constructed precisely so that it converges to a/n , the leading bias term itself, which is why subtracting it removes exactly that term. For a small number of classical statistics (including the variance estimator worked out in Example 2 below), the bias expansion

(1.6) is *exact* with no remainder at all, so the jackknife correction removes the bias completely rather than merely to a higher order.

Examples. A fully exact worked example (no remainder term at all) is given in Section 12, Example 2.

Applications. Bias correction of this kind is routinely used for ratio estimators in survey sampling, for maximum likelihood variance estimators, and for correlation and R^2 -type statistics in regression, all of which have bias of exactly the form (1.6) to leading order.

Mathematical Derivations

The Jackknife Standard Error Reduces to the Classical Formula for the Mean

For the sample mean, $\hat{\theta}_n = \bar{x} = n^{-1} \sum_i x_i$, the delete-one mean is $\hat{\theta}_{(i)} = (n\bar{x} - x_i)/(n-1)$, obtained by removing x_i from the total and dividing by the remaining $n-1$ observations. First,

$$\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)} = \frac{n\bar{x} - x_i}{n-1} - \bar{x} = \frac{n\bar{x} - x_i - (n-1)\bar{x}}{n-1} = \frac{\bar{x} - x_i}{n-1} = -\frac{x_i - \bar{x}}{n-1}, \quad (1.7)$$

using $\bar{\theta}_{(\cdot)} = \bar{x}$ exactly (an identity verified directly: $\sum_i \hat{\theta}_{(i)} = \sum_i (n\bar{x} - x_i)/(n-1) = (n^2\bar{x} - n\bar{x})/(n-1) = n\bar{x}$, so $\bar{\theta}_{(\cdot)} = \bar{x}$). Substituting (1.7) into the jackknife standard error formula (1.4),

$$\widehat{\text{se}}_{\text{jack}}^2 = \frac{n-1}{n} \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{(n-1)^2} = \frac{1}{n(n-1)} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{s^2}{n}, \quad (1.8)$$

where $s^2 = (n-1)^{-1} \sum_i (x_i - \bar{x})^2$ is the usual sample variance. Hence $\widehat{\text{se}}_{\text{jack}} = s/\sqrt{n}$ exactly reproduces the classical standard error formula for the mean — a useful sanity check confirming the jackknife formula (1.4) is correctly normalized.

Types of Bootstrap Confidence Intervals

Normal-approximation interval. Treating $\hat{\theta}$ as approximately Normally distributed with standard error $\widehat{\text{se}}_{\text{boot}}$ gives

$$\hat{\theta} \pm z_{1-\alpha/2} \widehat{\text{se}}_{\text{boot}}. \quad (1.9)$$

This is simple but ignores any skewness in the bootstrap distribution of $\hat{\theta}^*$.

Percentile interval. Using the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles of the bootstrap replicates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ directly,

$$(\hat{\theta}_{(\lceil B\alpha/2 \rceil)}^*, \hat{\theta}_{(\lceil B(1-\alpha/2) \rceil)}^*), \quad (1.10)$$

where $\hat{\theta}_{(k)}^*$ denotes the k -th order statistic of the bootstrap replicates. The percentile interval automatically adapts to skewness in the bootstrap distribution and is transformation-respecting (applying a monotone transformation to $\hat{\theta}$ and then computing a percentile interval gives the same interval, transformed, as computing the percentile interval first and then transforming its endpoints) — a property the normal-approximation interval lacks.

Algorithms

Algorithm 1.1: Bootstrap Estimation of Bias and Standard Error

Purpose. Estimate the bias and standard error of an arbitrary statistic without an analytical formula.

Input. Sample $x_1, \dots, x_n$; statistic function $T(\cdot)$; number of bootstrap replications B .

Output. $\widehat{\text{bias}}_{\text{boot}}, \widehat{\text{se}}_{\text{boot}}$.

Step-by-step procedure.

1. Compute $\hat{\theta} = T(x_1, \dots, x_n)$.
2. For $b = 1, \dots, B$: draw a bootstrap sample of size n with replacement from $\{x_1, \dots, x_n\}$; compute $\hat{\theta}_b^* = T(\cdot)$ on this resample.
3. Compute $\widehat{\text{bias}}_{\text{boot}}$ via (1.2) and $\widehat{\text{se}}_{\text{boot}}$ via (1.1).

Pseudo-code.

Algorithm: Bootstrap Bias and Standard Error

Input: $x[1..n]$, $T()$, B

Output: bias_boot , se_boot

1. $\text{theta_hat} \leftarrow T(x)$
2. For $b = 1$ to B :
 $x_star \leftarrow \text{sample}(x, \text{size} = n, \text{replace} = \text{TRUE})$
 $\text{theta_star}[b] \leftarrow T(x_star)$
3. $\text{bias_boot} \leftarrow \text{mean}(\text{theta_star}) - \text{theta_hat}$
4. $\text{se_boot} \leftarrow \text{sd}(\text{theta_star})$
5. Return bias_boot , se_boot

Flowchart description. Start; compute the statistic on the original sample; repeat B times: draw a with-replacement resample of size n , recompute the statistic; after the loop, compute the mean and standard deviation of the B replicates; report bias and standard error; end.

Complexity. $O(Bn)$ for the resampling itself, plus $O(B \cdot c(T))$ where $c(T)$ is the cost of one evaluation of the statistic.

Practical considerations. $B = 1000$ - 2000 is generally adequate for standard error estimation (Theorem 1.1); confidence intervals, especially percentile intervals, benefit from larger B (often $B \geq 2000$ - 5000) since they depend on tail quantiles rather than only the overall spread.

Algorithm 1.2: Bootstrap Percentile Confidence Interval

Purpose. Construct a confidence interval for θ that adapts to skewness in the bootstrap distribution.

Input. Sample $x_1, \dots, x_n$; statistic function $T(\cdot)$; number of replications B ; confidence level $1 - \alpha$.

Output. Interval $(\hat{\theta}_{\text{lower}}^*, \hat{\theta}_{\text{upper}}^*)$.

Step-by-step procedure.

1. Generate B bootstrap replicates $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$ as in Algorithm 1.1.
2. Sort the replicates and take the empirical $\alpha/2$ and $1 - \alpha/2$ quantiles as in (1.10).

Pseudo-code.

Algorithm: Bootstrap Percentile Interval

Input: $x[1..n]$, $T()$, B , α

Output: (lower, upper)

1. Generate $\text{theta_star}[1..B]$ as in Algorithm 1.1

```

2. lower <- quantile(theta_star, alpha / 2)
3. upper <- quantile(theta_star, 1 - alpha / 2)
4. Return (lower, upper)

```

Flowchart description. Start; generate B bootstrap replicates; sort them; read off the empirical quantiles at the two tail probabilities; report the resulting interval; end.

Complexity. $O(Bn + B \log B)$, the additional $B \log B$ coming from sorting the replicates.

Practical considerations. The percentile interval can perform poorly (be biased or too narrow) when the bootstrap distribution of $\hat{\theta}^*$ is itself substantially biased or skewed relative to θ ; the bias-corrected and accelerated (BCa) interval (Efron, 1987) addresses this at the cost of additional computation, and is recommended over the plain percentile interval whenever high accuracy is required.

Algorithm 1.3: Jackknife Estimation of Bias and Standard Error

Purpose. Estimate bias and standard error using n deterministic leave-one-out recomputations rather than random resampling.

Input. Sample $x_1, \dots, x_n$; statistic function $T(\cdot)$.

Output. $\widehat{\text{bias}}_{\text{jack}}, \widehat{\text{se}}_{\text{jack}}$.

Step-by-step procedure.

1. Compute $\hat{\theta} = T(x_1, \dots, x_n)$.
2. For $i = 1, \dots, n$: compute $\hat{\theta}_{(i)} = T(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$.
3. Compute $\bar{\theta}_{(\cdot)}$, then $\widehat{\text{bias}}_{\text{jack}}$ via (1.3) and $\widehat{\text{se}}_{\text{jack}}$ via (1.4).

Pseudo-code.

Algorithm: Jackknife Bias and Standard Error

Input: $x[1..n]$, $T()$

Output: bias_jack , se_jack

```

1. theta_hat <- T(x)
2. For i = 1 to n:
    theta_i[i] <- T(x with element i removed)
3. theta_bar <- mean(theta_i)
4. bias_jack <- (n - 1) * (theta_bar - theta_hat)
5. se_jack <- sqrt((n - 1) / n * sum((theta_i - theta_bar)^2))
6. Return bias_jack, se_jack

```

Flowchart description. Start; compute the statistic on the full sample; for each observation in turn, remove it and recompute the statistic; after all n recomputations, compute their mean and sum of squared deviations; report bias and standard error; end.

Complexity. $O(n \cdot c(T))$, exactly n evaluations of the statistic, with no random resampling required.

Practical considerations. The jackknife can fail for non-smooth statistics — most notoriously the sample median, for which the jackknife standard error estimate is inconsistent (it does not converge to the correct value even as $n \rightarrow \infty$); the bootstrap does not share this defect and is generally preferred for such statistics.

R Programming

```

set.seed(123)

#' Bootstrap estimates of bias and standard error

```

```

#'
#' Purpose: implements Algorithm 1.1.
#'
#' Arguments:
#'   sample_data      numeric vector, the observed sample
#'   statistic_function function mapping a numeric vector to a scalar
#'   number_of_replicates number of bootstrap resamples (default 2000)
#'
#' Returns:
#'   a list with the original estimate, bootstrap bias, bootstrap
#'   standard error, and the full vector of bootstrap replicates
#'
#' Example:
#'   bootstrap_estimate(rnorm(30), mean, number_of_replicates = 1000)
bootstrap_estimate <- function(sample_data, statistic_function, number_of_replicates = 2000) {
  sample_size <- length(sample_data)
  original_estimate <- statistic_function(sample_data)

  bootstrap_replicates <- numeric(number_of_replicates)
  for (replicate_index in seq_len(number_of_replicates)) {
    resampled_data <- sample(sample_data, size = sample_size, replace = TRUE)
    bootstrap_replicates[replicate_index] <- statistic_function(resampled_data)
  }

  list(
    estimate = original_estimate,
    bootstrap_bias = mean(bootstrap_replicates) - original_estimate,
    bootstrap_se = sd(bootstrap_replicates),
    replicates = bootstrap_replicates
  )
}

#' Bootstrap percentile confidence interval
#'
#' Purpose: implements Algorithm 1.2.
#'
#' Arguments:
#'   sample_data      numeric vector, the observed sample
#'   statistic_function function mapping a numeric vector to a scalar
#'   number_of_replicates number of bootstrap resamples (default 2000)
#'   confidence_level  confidence level (default 0.95)
#'
#' Returns:
#'   a named numeric vector with the lower and upper interval endpoints
#'
#' Example:
#'   bootstrap_percentile_ci(rnorm(30), mean)
bootstrap_percentile_ci <- function(sample_data, statistic_function,
                                   number_of_replicates = 2000, confidence_level = 0.95) {
  boot_result <- bootstrap_estimate(sample_data, statistic_function, number_of_replicates)
  alpha <- 1 - confidence_level

  quantile(boot_result$replicates, probs = c(alpha / 2, 1 - alpha / 2))
}

```

```

}

#' Jackknife estimates of bias and standard error
#'
#' Purpose: implements Algorithm 1.3.
#'
#' Arguments:
#'   sample_data      numeric vector, the observed sample
#'   statistic_function function mapping a numeric vector to a scalar
#'
#' Returns:
#'   a list with the original estimate, jackknife bias, jackknife
#'   standard error, the bias-corrected estimate, and the vector of
#'   leave-one-out replicates
#'
#' Example:
#'   jackknife_estimate(rnorm(30), mean)
jackknife_estimate <- function(sample_data, statistic_function) {
  sample_size <- length(sample_data)
  original_estimate <- statistic_function(sample_data)

  leave_one_out_replicates <- numeric(sample_size)
  for (index in seq_len(sample_size)) {
    leave_one_out_replicates[index] <- statistic_function(sample_data[-index])
  }

  mean_leave_one_out <- mean(leave_one_out_replicates)
  jackknife_bias <- (sample_size - 1) * (mean_leave_one_out - original_estimate)
  jackknife_se <- sqrt(
    (sample_size - 1) / sample_size *
    sum((leave_one_out_replicates - mean_leave_one_out)^2)
  )

  list(
    estimate = original_estimate,
    jackknife_bias = jackknife_bias,
    jackknife_se = jackknife_se,
    bias_corrected_estimate = original_estimate - jackknife_bias,
    replicates = leave_one_out_replicates
  )
}

```

Simulation Studies

Bootstrap Standard Error Convergence (Theorem 1.1)

```

set.seed(123)

fixed_sample <- rnorm(30, mean = 5, sd = 2)
true_theoretical_se <- sd(fixed_sample) / sqrt(length(fixed_sample))

replication_counts <- c(10, 50, 200, 500, 1000, 5000, 20000)

```

```
se_estimates <- vapply(
  replication_counts,
  function(B) bootstrap_estimate(fixed_sample, mean, number_of_replicates = B)$bootstrap_se,
  numeric(1)
)

tibble(B = replication_counts, Bootstrap_SE = se_estimates, Classical_SE = true_theoretical_se) %>%
  kable(digits = 4, caption = "Bootstrap SE of the mean converges to the classical formula s/sqrt(n) as
```

Table 1: Bootstrap SE of the mean converges to the classical formula $s/\sqrt{n}$ as B grows

B	Bootstrap_SE	Classical_SE
10	0.4357	0.3582
50	0.4031	0.3582
200	0.3529	0.3582
500	0.3626	0.3582
1000	0.3551	0.3582
5000	0.3515	0.3582
20000	0.3509	0.3582

Jackknife Bias Correction for the Variance Estimator

For $X_1, \dots, X_n$ i.i.d. $\text{Normal}(\mu, \sigma^2)$, the plug-in (divide-by- n) variance estimator $\hat{\sigma}_n^2 = n^{-1} \sum_i (X_i - \bar{X})^2$ has the exact bias $E(\hat{\sigma}_n^2) - \sigma^2 = -\sigma^2/n$, matching the form of (1.6) exactly with $a = -\sigma^2$ and no remainder term.

```
set.seed(123)

plugin_variance <- function(x) mean((x - mean(x))^2)

true_sigma_squared <- 4
number_of_datasets <- 3000
sample_size_n <- 15

bias_estimates <- replicate(number_of_datasets, {
  simulated_sample <- rnorm(sample_size_n, mean = 0, sd = sqrt(true_sigma_squared))
  jackknife_estimate(simulated_sample, plugin_variance)$jackknife_bias
})

tibble(
  Quantity = c("Theoretical bias (-sigma^2 / n)", "Average jackknife bias estimate"),
  Value = c(-true_sigma_squared / sample_size_n, mean(bias_estimates))
) %>%
  kable(digits = 4, caption = "Jackknife bias estimate vs. the exact theoretical bias of the plug-in va
```

Table 2: Jackknife bias estimate vs. the exact theoretical bias of the plug-in variance estimator

Quantity	Value
Theoretical bias (-sigma^2 / n)	-0.2667
Average jackknife bias estimate	-0.2671

Coverage of the Bootstrap Percentile Confidence Interval

```
set.seed(123)

number_of_trials <- 500
sample_size_coverage <- 25
true_mean <- 10
coverage_indicator <- logical(number_of_trials)

for (trial in seq_len(number_of_trials)) {
  trial_sample <- rnorm(sample_size_coverage, mean = true_mean, sd = 3)
  interval <- bootstrap_percentile_ci(trial_sample, mean, number_of_replicates = 500)
  coverage_indicator[trial] <- (interval[1] <= true_mean) && (true_mean <= interval[2])
}

tibble(
  Quantity = c("Nominal coverage", "Empirical coverage (500 trials)"),
  Value = c(0.95, mean(coverage_indicator))
) %>%
  kable(digits = 4, caption = "Empirical coverage of the 95% bootstrap percentile interval for the mean")
```

Table 3: Empirical coverage of the 95% bootstrap percentile interval for the mean

Quantity	Value
Nominal coverage	0.950
Empirical coverage (500 trials)	0.936

Graphs

```
set.seed(123)

boot_result_plot <- bootstrap_estimate(fixed_sample, mean, number_of_replicates = 3000)
normal_ci <- boot_result_plot$estimate + c(-1, 1) * qnorm(0.975) * boot_result_plot$bootstrap_se
percentile_ci <- quantile(boot_result_plot$replicates, probs = c(0.025, 0.975))

ggplot(tibble(theta_star = boot_result_plot$replicates), aes(x = theta_star)) +
  geom_histogram(aes(y = after_stat(density)), bins = 40, fill = "steelblue", colour = "white") +
  geom_vline(xintercept = normal_ci, colour = "darkred", linetype = "dashed", linewidth = 0.8) +
  geom_vline(xintercept = percentile_ci, colour = "darkgreen", linetype = "dotted", linewidth = 0.8) +
  labs(
    title = "Bootstrap distribution of the sample mean (n = 30)",
    subtitle = "Dashed red = normal-approximation CI; dotted green = percentile CI",
    x = expression(hat(theta)^{"*"}), y = "Density"
  ) +
  theme_minimal(base_size = 11)

illustrative_data <- c(5, 7, 9, 12, 14)
jackknife_illustrative <- jackknife_estimate(illustrative_data, mean)

ggplot(
  tibble(observation_removed = illustrative_data, leave_one_out_mean = jackknife_illustrative$replicates)
```

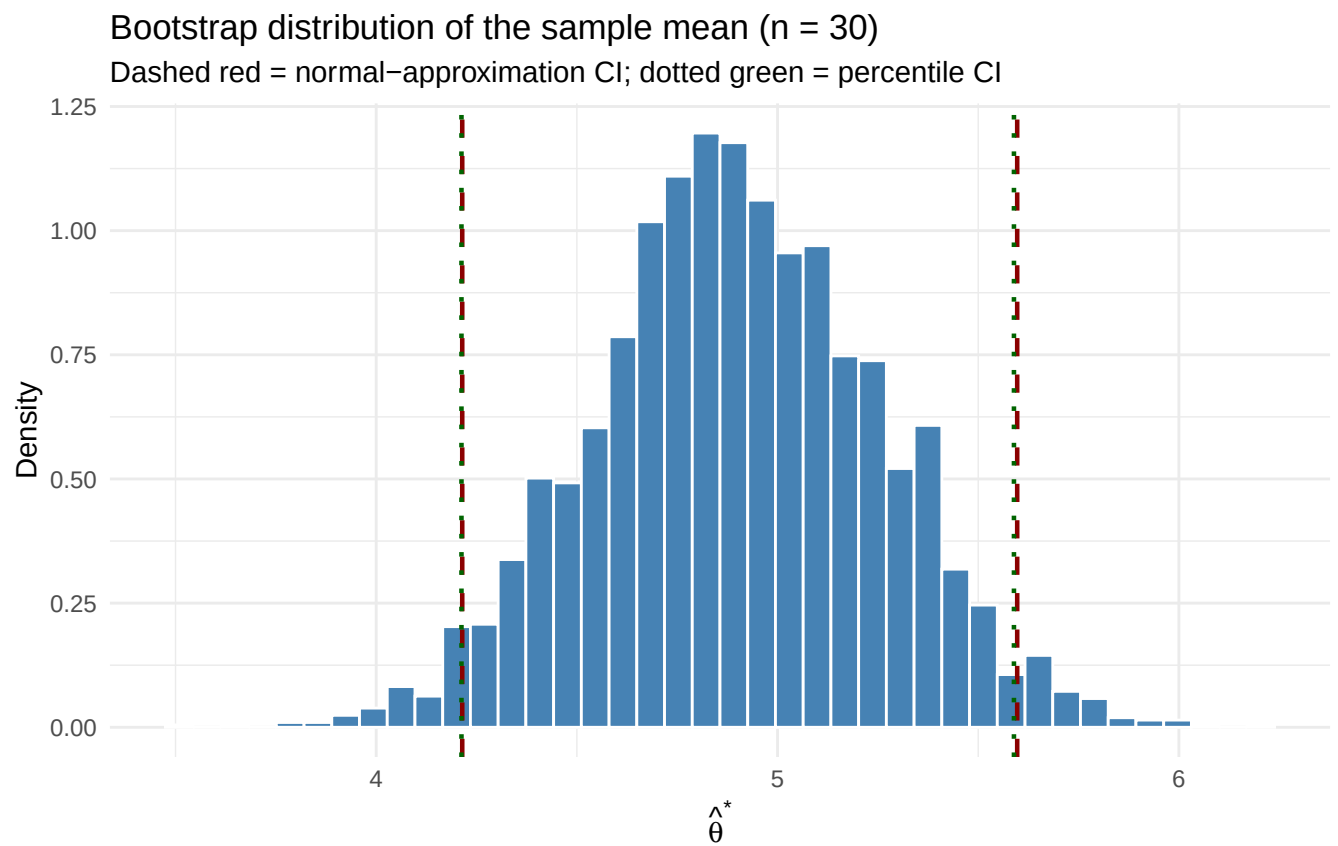


Figure 2: Bootstrap distribution of the sample mean, with the normal-approximation and percentile 95% intervals marked.

```

aes(x = factor(observation_removed), y = leave_one_out_mean)
) +
geom_col(fill = "darkorange", width = 0.6) +
geom_hline(yintercept = jackknife_illustrative$estimate, colour = "black", linetype = "dashed") +
labs(
  title = "Leave-one-out sample means",
  x = "Observation removed", y = "Mean of remaining 4 observations"
) +
theme_minimal(base_size = 11)

```

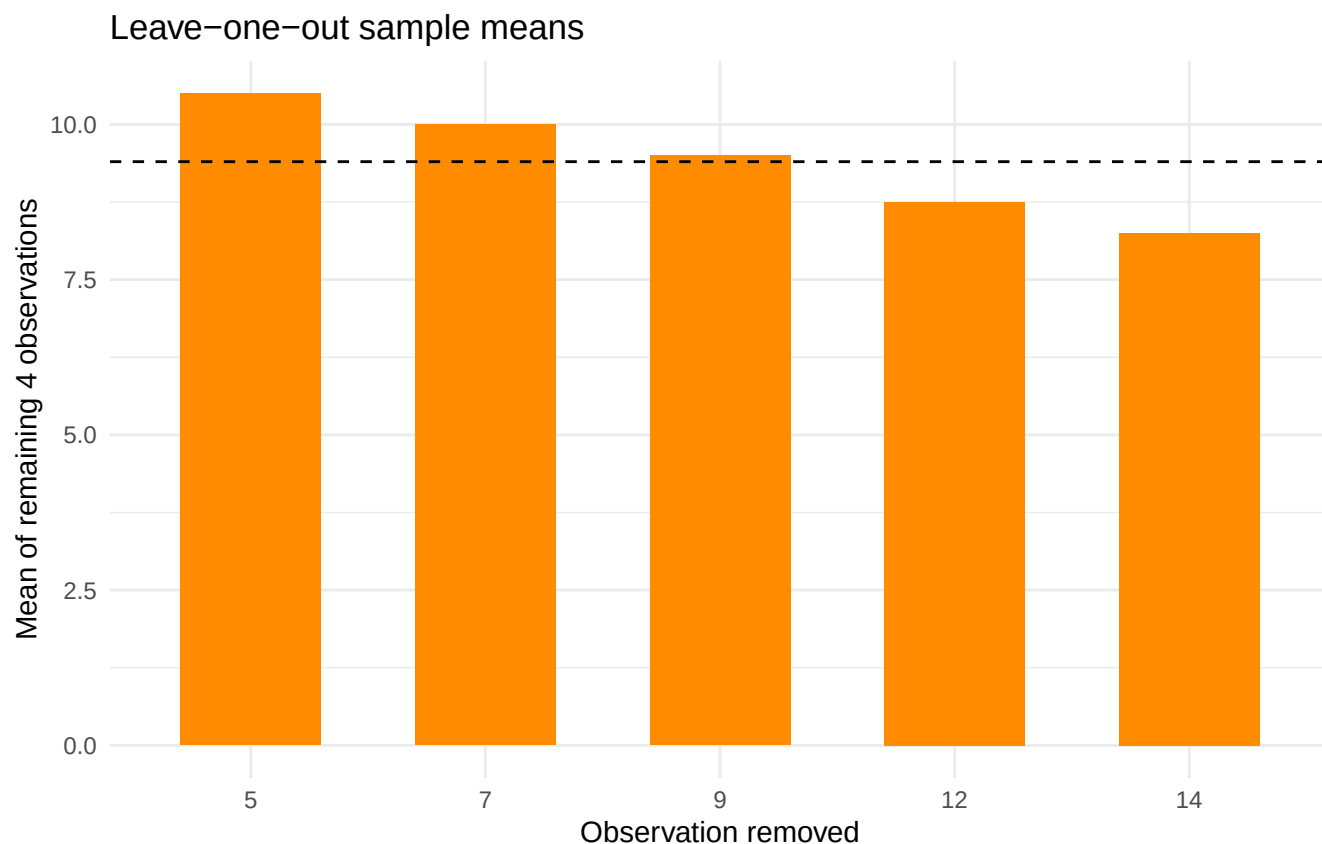


Figure 3: Leave-one-out means for the illustrative dataset of Section 12, showing the influence of each observation.

Numerical Examples

Example 1: Bootstrap and jackknife standard error of the mean.

Problem. For the sample $x = (5, 7, 9, 12, 14)$ ($n = 5$), compute the classical, jackknife, and (conceptually) bootstrap standard errors of the sample mean.

Solution. $\bar{x} = 47/5 = 9.4$. The sample variance is $s^2 = 53.2/4 = 13.3$, giving the classical standard error $s/\sqrt{5} = 3.6469/2.2361 = 1.6309$.

For the jackknife, the leave-one-out means are 10.5, 10.0, 9.5, 8.75, 8.25 (removing 5, 7, 9, 12, 14 respectively), with $\bar{\theta}_{(\cdot)} = 9.4$ (exactly $\bar{x}$, as shown in Section 8.1). The sum of squared deviations of the leave-one-out means from 9.4 is $1.1^2 + 0.6^2 + 0.1^2 + (-0.65)^2 + (-1.15)^2 = 3.325$, so

$$\widehat{\text{se}}_{\text{jack}} = \sqrt{\frac{4}{5} \times 3.325} = \sqrt{2.66} = 1.6310,$$

matching the classical value to rounding error, exactly as (1.8) guarantees.

Interpretation. This example both illustrates the mechanics of the jackknife and validates it: because the sample mean has a known closed-form standard error, computing the jackknife estimate by hand and confirming it reproduces $s/\sqrt{n}$ builds confidence in the general formula before applying it to statistics without a known closed form.

Example 2: Exact jackknife bias correction of the plug-in variance.

Problem. For the same sample, compute the plug-in (divide-by- n) variance estimate $\hat{\sigma}_n^2$, its jackknife bias estimate, and the bias-corrected estimator, and compare to the classical divide-by- $(n-1)$ estimator.

Solution. $\hat{\sigma}_n^2 = 53.2/5 = 10.64$. The five leave-one-out plug-in variances (computed on the four remaining points each time) are 7.25, 11.5, 13.25, 11.1875, 6.6875, averaging to $\bar{\theta}_{(\cdot)} = 9.975$. Hence

$$\widehat{\text{bias}}_{\text{jack}} = 4 \times (9.975 - 10.64) = -2.66,$$

and the bias-corrected estimator is $\hat{\theta}_{\text{jack}} = 10.64 - (-2.66) = 13.30$, which is *exactly* s^2 , the classical unbiased variance estimator computed in Example 1.

Interpretation. For the variance statistic, the jackknife bias correction is not merely an asymptotic $O(1/n^2)$ improvement (Theorem 1.2); it recovers the exactly unbiased estimator with no remainder at all, because the bias of the plug-in variance estimator is *exactly* $-\sigma^2/n$ with no higher-order terms — this is the special case referenced in the Remarks following Theorem 1.2.

Applications

- **Statistics.** Bootstrap confidence intervals are now standard for regression coefficients, correlation and R^2 statistics, and any estimator lacking a tractable analytical standard error.
- **Machine Learning.** Bootstrap resampling underlies bagging (bootstrap aggregating) and random forests directly, and is widely used to construct confidence intervals for cross-validated performance metrics.
- **Data Science.** Quantifying uncertainty in ad hoc summary statistics (medians, quantiles, custom business metrics) computed on observational data, where no analytical formula is available.
- **Finance.** Bootstrap resampling of historical returns is used to estimate the sampling variability of portfolio risk measures without assuming a specific parametric return distribution.
- **Reliability Engineering.** Jackknife and bootstrap bias correction for reliability function estimators computed from censored failure time data.
- **Monte Carlo Methods.** The bootstrap is itself a Monte Carlo technique, connecting directly to the simulation methods developed throughout Unit 3.
- **Bayesian Statistics.** The bootstrap has a Bayesian analogue (the Bayesian bootstrap, Rubin 1981) and bootstrap ideas inform the construction of approximate posterior predictive checks.
- **Operations Research.** Bootstrap confidence intervals for simulation output statistics (e.g. mean queue length) when the underlying distribution of simulation replications is not Normal.
- **Engineering.** Jackknife bias correction of parameter estimates in fitted degradation or fatigue models with small sample sizes.

Advantages

- Both methods require no analytical derivation of a standard error formula; they apply to essentially any statistic that can be computed on a data vector.

- The bootstrap directly approximates the full sampling *distribution* of a statistic, not merely its standard error, enabling confidence intervals that adapt to skewness (the percentile method).
- The jackknife requires only n deterministic recomputations, with no random number generation needed, and its bias-correction identity is exact for several classical statistics.

Limitations

- The bootstrap's accuracy depends on how well the empirical distribution $\hat{F}$ approximates the true population F , which can be poor for small n or for statistics sensitive to the tails of the distribution.
- The jackknife is inconsistent for non-smooth statistics, most notably the sample median, and should not be used for standard error estimation in that setting.
- Both methods can be computationally expensive when the statistic itself is costly to compute (for example, refitting a complex model for each of thousands of bootstrap resamples).
- Neither method rescues a fundamentally biased sampling design; resampling reflects the properties of the *observed* sample, and cannot correct for problems such as non-random sampling from the population of interest.

Practical Tips

- **Use at least $B = 1000$ -2000 replications for standard error estimation, and more for confidence intervals.** As Theorem 1.1 and the simulation in Section 10.1 show, the bootstrap standard error estimate itself has Monte Carlo error that shrinks only as $B \rightarrow \infty$; percentile intervals depend on tail quantiles and typically need larger B for the same stability.
- **Always set a random seed for bootstrap reproducibility**, since each run otherwise produces a slightly different result.
- **Do not apply the plain jackknife to the sample median or other non-smooth statistics.** Use the bootstrap instead, or a smoothed jackknife variant if the jackknife's computational economy is essential.
- **Check whether the bootstrap distribution is markedly skewed before relying on the normal-approximation interval.** The percentile interval (or a more refined method such as BCa) is generally more trustworthy when skewness is present.
- **Vectorize where possible.** As implemented above, resampling and recomputing a simple statistic like the mean is fast even at large B ; for computationally expensive statistics, consider reducing B or parallelizing the replicate loop.

Summary

- The bootstrap approximates the sampling distribution of a statistic by resampling with replacement from the observed data, treating the empirical distribution as a stand-in for the unknown population.
- Bootstrap estimates of bias and standard error are computed directly from the spread of bootstrap replicates (Theorem 1.1 establishes their Monte Carlo consistency as $B \rightarrow \infty$).
- Bootstrap confidence intervals can be constructed via the normal approximation or, more robustly to skewness, via the percentile method.
- The jackknife estimates bias and standard error using n systematic leave-one-out recomputations; Theorem 1.2 shows it removes the leading $O(1/n)$ bias term for smooth statistics, and this correction is exact (not merely asymptotic) for the plug-in variance estimator, as Example 2 demonstrates.
- The jackknife's simplicity comes at the cost of inconsistency for non-smooth statistics such as the median, where the bootstrap remains the appropriate tool.

Key Formulae

- Bootstrap SE: $\widehat{\text{se}}_{\text{boot}} = \text{sd}(\hat{\theta}_1^*, \dots, \hat{\theta}_B^*)$.

- Bootstrap bias: $\widehat{\text{bias}}_{\text{boot}} = \bar{\theta}^* - \hat{\theta}$.
- Bootstrap percentile CI: empirical $(\alpha/2, 1 - \alpha/2)$ quantiles of $\hat{\theta}_1^*, \dots, \hat{\theta}_B^*$.
- Jackknife bias: $\widehat{\text{bias}}_{\text{jack}} = (n-1)(\bar{\theta}_{(\cdot)} - \hat{\theta})$.
- Jackknife SE: $\widehat{\text{se}}_{\text{jack}} = \sqrt{\frac{n-1}{n} \sum_i (\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)})^2}$.
- Jackknife bias-corrected estimator: $\hat{\theta}_{\text{jack}} = n\hat{\theta} - (n-1)\bar{\theta}_{(\cdot)}$.

Key Terms

- **Bootstrap principle.** Approximating the sampling distribution of a statistic by resampling from the empirical distribution.
- **Bootstrap replicate.** The value of a statistic computed on one bootstrap resample.
- **Empirical distribution.** The distribution placing probability $1/n$ on each observed data point.
- **Jackknife.** A resampling technique based on systematically leaving out one observation at a time.
- **Leave-one-out statistic.** The value of a statistic recomputed with a single observation removed.
- **Percentile interval.** A bootstrap confidence interval formed from the empirical quantiles of the bootstrap replicates.
- **Resampling.** Generating new samples from an observed dataset, either with replacement (bootstrap) or by systematic deletion (jackknife), to approximate sampling variability.

Exercises

Multiple Choice Questions

1. A bootstrap sample is drawn
 - (a) without replacement, of size $n - 1$
 - (b) with replacement, of size n
 - (c) without replacement, of size n
 - (d) with replacement, of size $2n$
2. The bootstrap estimate of bias is
 - (a) $\hat{\theta} - \bar{\theta}^*$
 - (b) $\bar{\theta}^* - \hat{\theta}$
 - (c) $\text{sd}(\hat{\theta}^*)$
 - (d) $\hat{\theta}^2$
3. As $B \rightarrow \infty$, the bootstrap standard error estimate converges to
 - (a) the true population standard error
 - (b) zero
 - (c) the “ideal bootstrap” standard error under resampling from $\hat{F}$
 - (d) infinity
4. The jackknife leave-one-out statistic $\hat{\theta}_{(i)}$ is computed
 - (a) on a bootstrap resample of size n
 - (b) on the original sample with observation i removed
 - (c) on a random subsample of size $n/2$
 - (d) on the full sample with observation i duplicated
5. For the sample mean, the jackknife standard error
 - (a) is inconsistent
 - (b) reduces exactly to $s/\sqrt{n}$
 - (c) is always larger than $s/\sqrt{n}$
 - (d) cannot be computed
6. The jackknife bias correction, under the expansion (1.6), reduces the order of the bias from
 - (a) $O(1/n)$ to $O(1)$
 - (b) $O(1/n^2)$ to $O(1/n)$
 - (c) $O(1/n)$ to $O(1/n^2)$

- (d) $O(1)$ to $O(1/n)$
- 7. The jackknife is known to be inconsistent for
 - (a) the sample mean
 - (b) the sample variance
 - (c) the sample median
 - (d) the sample sum
- 8. The bootstrap percentile confidence interval is constructed from
 - (a) the mean and standard deviation of the bootstrap replicates assuming normality
 - (b) the empirical quantiles of the bootstrap replicates
 - (c) the leave-one-out replicates only
 - (d) the original sample alone
- 9. For the plug-in (divide-by- n) variance estimator under Normal sampling, the exact bias is
 - (a) 0
 - (b) σ^2/n
 - (c) $-\sigma^2/n$
 - (d) $-\sigma^2/(n-1)$
- 10. Compared to the jackknife, the bootstrap generally requires
 - (a) fewer recomputations of the statistic
 - (b) exactly n recomputations
 - (c) more recomputations of the statistic, but random resampling rather than systematic deletion
 - (d) no recomputation at all

Answers. 1(b), 2(b), 3(c), 4(b), 5(b), 6(c), 7(c), 8(b), 9(c), 10(c).

Short Answer Questions

1. State the bootstrap principle in your own words.
2. Define the empirical distribution $\hat{F}$ and explain its role in the bootstrap.
3. Write down the formulas for the bootstrap estimates of bias and standard error.
4. Write down the formulas for the jackknife estimates of bias and standard error.
5. Explain why the jackknife standard error formula includes the factor $(n-1)/n$ rather than simply $1/n$.
6. Explain, using Theorem 1.1, what aspect of bootstrap accuracy is controlled by increasing B , and what aspect is not.
7. Why is the percentile confidence interval generally preferred over the normal-approximation interval when the bootstrap distribution is skewed?
8. Explain why the jackknife is inconsistent for the sample median.
9. State the jackknife bias-correction identity and explain what “removing the leading bias term” means.
10. Give one practical application, distinct from those emphasized in this unit, where bootstrap resampling would be useful.

Long Answer Questions

1. State and prove Theorem 1.1 (consistency of the bootstrap standard error estimate as $B \rightarrow \infty$), and explain what source of error this theorem does and does not address.
2. State and prove Theorem 1.2 (jackknife bias reduction), explaining the role of the assumed bias expansion (1.6).
3. Derive the identity showing that the jackknife standard error reduces exactly to $s/\sqrt{n}$ for the sample mean, as in Section 8.1.
4. Using the dataset of Example 2, explain why the jackknife bias correction of the plug-in variance estimator is exact rather than merely asymptotic, and relate this to the general theory of Theorem 1.2.
5. Compare the bootstrap and the jackknife in terms of computational cost, applicability to non-smooth statistics, and the type of inferential statement (standard error only, versus full sampling distribution) each supports.
6. Discuss the normal-approximation and percentile bootstrap confidence intervals, explaining under what circumstances they would give noticeably different results.

7. Explain the connection between the bootstrap and the general Monte Carlo simulation framework developed in Unit 3, identifying which specific generation method (from Units 3.1-3.6) underlies drawing a bootstrap resample.
8. Discuss the role of resampling methods in modern machine learning, with specific reference to bagging and bootstrap-based confidence intervals for model performance metrics.
9. Explain why neither the bootstrap nor the jackknife can correct for a biased sampling design, using a concrete hypothetical example.
10. Discuss the historical development of the jackknife (Quenouille, Tukey) and the bootstrap (Efron), and explain the conceptual relationship between the two methods.

Programming Exercises

1. Implement `bootstrap_estimate()` from first principles (already provided) and reproduce, by simulation, the standard error comparison of Example 1 for the dataset given there.
2. Implement `jackknife_estimate()` from first principles (already provided) and reproduce the manual calculations of Examples 1 and 2 exactly.
3. Apply the bootstrap to estimate the standard error of the sample correlation coefficient for a bivariate dataset of your choosing (for example, simulated from a bivariate Normal distribution with a known correlation), and compare the bootstrap standard error to the asymptotic formula $se(\hat{\rho}) \approx (1 - \rho^2)/\sqrt{n}$.
4. Demonstrate empirically that the jackknife standard error is inconsistent for the sample median: for increasing sample sizes drawn from a known continuous distribution, compare the jackknife standard error of the median to a bootstrap estimate, and discuss the discrepancy.
5. Construct and compare the normal-approximation and percentile 95% bootstrap confidence intervals for the coefficient of variation ($s/\bar{x}$) of a right-skewed simulated dataset (e.g. drawn from a Gamma or Log-normal distribution), and discuss which interval appears more trustworthy.
6. Write a simulation study estimating the empirical coverage of the normal-approximation bootstrap interval, analogous to Section 10.3, and compare its coverage to that of the percentile interval under the same conditions.
7. Implement the jackknife bias correction for the plug-in variance estimator for a range of sample sizes $n \in \{5, 10, 20, 50\}$, and verify, by simulation, that the corrected estimator's average bias is negligible in every case (since the correction is exact for this statistic).
8. Apply bootstrap resampling to estimate the standard error of a fitted simple linear regression slope coefficient, using case resampling (resampling entire (x_i, y_i) pairs), and compare to the classical formula-based standard error from `lm()`.
9. Benchmark, using `system.time()`, the bootstrap (with $B = 2000$) against the jackknife for a sample of size $n = 200$ and a moderately expensive statistic (for example, a trimmed mean), and discuss the relative computational cost.
10. Implement a simple Bayesian bootstrap (Rubin, 1981): instead of resampling with replacement, assign each observation a random Dirichlet($1, \dots, 1$) weight and compute a weighted statistic; compare the resulting distribution of replicates to the ordinary bootstrap distribution for the sample mean.

Mini Projects

1. **A resampling toolkit with diagnostics.** Build a small R toolkit that, given a dataset and a statistic function, automatically computes the bootstrap bias, bootstrap standard error, jackknife bias, jackknife standard error, and both types of bootstrap confidence interval, and produces a diagnostic report flagging (i) substantial disagreement between the bootstrap and jackknife standard errors, and (ii) substantial skewness in the bootstrap distribution (for example, via a simple skewness statistic). Demonstrate the toolkit on at least three different statistics (mean, variance, and one statistic of your choosing that is plausibly non-smooth, such as the median), and discuss what the diagnostics reveal in each case.
2. **Bootstrap versus jackknife under model misspecification.** Design a simulation study in which data are generated from a heavy-tailed distribution (for example, a t_3 distribution), and compare the bootstrap and jackknife standard error estimates for the sample mean and the 10% trimmed mean across

repeated simulated datasets, evaluating each against the true (simulation-based) sampling standard error. Discuss which method degrades more gracefully as the tail heaviness increases, and relate your findings to the smoothness assumptions underlying Theorem 1.2.

Difficulty labels. Multiple choice, short answer: **Easy**. Long answer questions 3, 4, 6, 7, 9, programming exercises 1-6: **Medium**. Long answer questions 1, 2, 5, 8, 10, programming exercises 7-10: **Medium to Advanced**. Mini projects: **Advanced**.

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